STARSHIP OPERATIONS

The stars await! Adventurers ply the star-lanes and boldly venture into the unknown; others may serve in a navy, hunting pirates, or raiding enemy shipping. Still others may mine asteroid belts for valuable ores, smuggle goods to distant planets, or even engage in piracy. This chapter covers interplanetary and interstellar travel, as well as starship operations and procedures.

INTERPLANETARY TRAVEL

Starships and other spacecraft in Cepheus Deluxe have reactionless real-space thrusters with an acceleration rating from 1G to 6Gs. Ships normally travel under full acceleration halfway to their destination, then decelerate for the other half of the journey. The following table presents abstracted in-system travel times according to the ship's acceleration.

Travel times are abstracted and, as noted above, generally assume constant acceleration up to the halfway point and then constant deceleration down to the destination. Ships intending to jump will not decelerate towards the jump point. They will emerge from a jump at the speed they entered it.

Common interplanetary travel times by acceleration

Destination	1-G	2-G	3-G	4-G	5-G	6G
Planetoid Jump Point	10m	7m	6m	5m	5m	4m
Ground to Planetary Orbit	33m	23m	19m	16m	14m	13m
Ground to Typical Planetary Jump Point	5h, 30m	4h	3h	3h	2h, 30m	2h
Small Gas Giant to Jump Point	12h	8h	7h	6h	5h, 30m	5h
Large Gas Giant to Jump Point	17h	12h	10h	9h	8h	7h
Planet to Close Neighbor	1d, 13h	1d, 2h	22h	19h	17h	15h
Planet to Far Neighbor	3d, 17h	2d, 15h	2d, 3h	1d, 20h	1d, 16h	1d, 12h
Mainworld to Close Gas Giant	5d, 16h	4d	3d, 7h	2d, 20h	2d, 13h	2d, 8h
Mainworld to Far Gas Giant	6d, 23h	4d, 22h	4d	3d, 11h	3d, 3h	2d, 20h



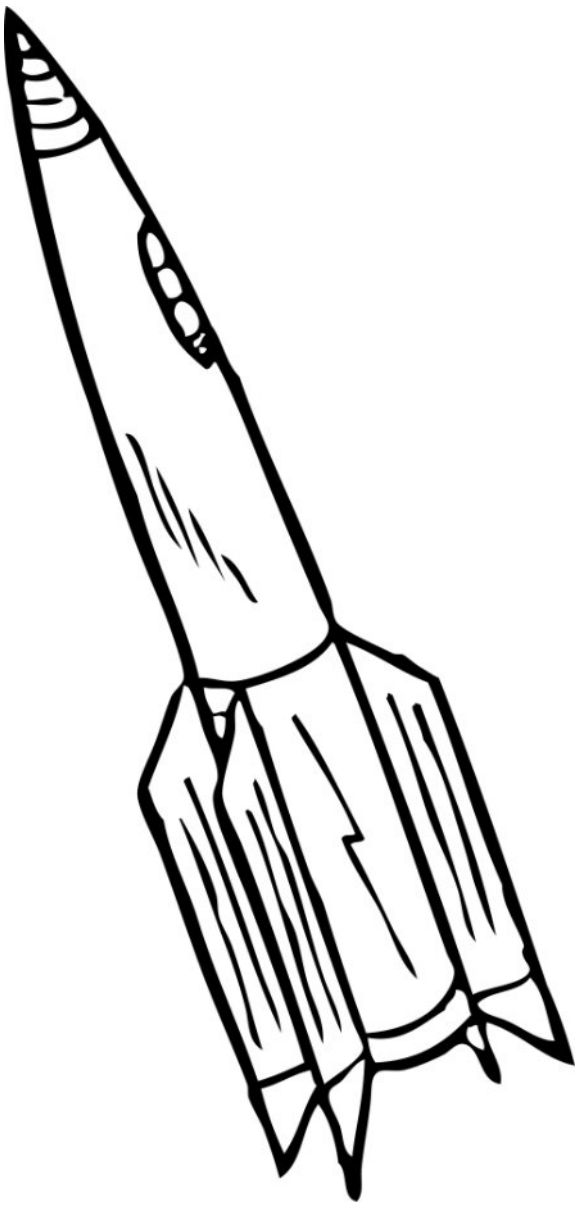
INTERSTELLAR TRAVEL

Interstellar travel in Cepheus Deluxe uses a faster-than-light engine called the Jump Drive. Jump Drives are rated according to the maximum range in parsecs (represented on the star map as hexagons) they may travel in a single jump. Jump-1 drives can propel a vessel to an adjacent hex, while Jump-3 drives will move a ship up to three hexes. A single voyage through jump-space takes 1 week regardless of the distance traveled. Jump drives require a significant gravity well, like a star system, at both ends of the jump, but not too much gravity. It is impossible to purposefully jump to a destination without a star system. Jump drives are safe to use only at a certain distance from a planetary or stellar body. This is called the Jump Point or Jump Distance. A jump requires an amount of liquid hydrogen fuel equal to 10% of the ship's volume per parsec traveled. For example, a 200-ton ship making a 3-parsec jump would burn 60 tons of fuel.

The jump drive propels the ship into a parallel dimension, called Jump Space. The jump fuel serves to envelope the ship with a hydrogen bubble, at a distance of approximately 5m in every direction from the hull – the so-called "Jump Bubble" – to protect it from the alien nature of Jump Space. Characters can safely go on EVA within the bubble but will be lost if they leave it and enter real, unprotected Jump Space – even if still tethered to the ship.

INTERSTELLAR JUMP PROCEDURE

1. **Calculate Plot:** A jump requires a course calculation called a "jump plot". Starports rated D or better sell up-to-date jump plots to nearby populated worlds for Cr1000 per parsec. A skilled pilot can also calculate their own jump plot. Throw Piloting/EDU 6+ to calculate a jump plot, apply a negative DM equal to the number of parsecs the jump will travel; this takes 1Dx6 minutes (i.e., 1D ship combat turns). It is possible to rush this calculation, to guarantee that it will only take one ship combat turn, but the throw is Piloting/EDU 10+. A failed check requires another attempt, or the ship will misjump (see below). It is possible to calculate a jump plot in advance, but it will become obsolete within 1D days.
2. **Engage Drive:** Once the jump plot is ready, the ship's engineer must throw Engineering/EDU 8+ to engage the Jump Drive. This procedure takes a second and can be done in the same round as the jump throw itself.
3. **Jump!:** To jump, throw 2D with the following DMs. A throw of 0 or less causes a misjump. A bad jump plot causes an automatic misjump.
    1. DM+the Effect of the Engineering throw to Engage the Drive
    2. DM-1 per day the jump plot is out of date
    3. DM-2 if the Jump Drive is damaged.
    4. DM-2 if using Unrefined fuel.
    5. DM-8 if within a hundred-diameter limit of the nearest planet or star (Not at a jump point)
4. **In Case of a Misjump:** A misjump causes a Critical Hit on the ship (see the space combat rules) and may destroy it. Furthermore, the ship travels 1D x 1D parsecs in a random direction. If this leads to an empty hex on the map, the ship emerges from the misjump at the closest star system to that empty hex. In case there are multiple adjacent systems, choose one randomly.



STARSHIP EXPENSES

A starship's owner must handle several expenses, detailed below.

MORTGAGE

A mortgaged ship's owner must pay 1/320th of the ship's cash price per month for 480 months (40 years). This means that the owner pays 150% of the ship's purchase price back to the bank.

CREW SALARIES

A ship needs the following crew-members:

Captain: Cr6000 per month if hired (i.e., when the captain is not the ship's owner/operator). The captain leads the crew and manages the ship and requires at least Leadership 1. Any crew member can serve as captain in addition to another job (at no additional pay), but this limits their capability to lead the crew in battle or other emergency. Furthermore, a ship can have a captain without the Leadership skill, but such captain will be unable to use the special captain actions described in the Space Combat chapter. A dedicated, skilled captain is essential if the ship expects combat.

Pilot: Cr6000 per month. Requires at least Piloting 1. The Pilot flies and maneuvers the ship, as well as plots interplanetary movement and interstellar jumps. Every ship requires a pilot. Military ships will have more than one pilot in the event of crew casualties.

Sensor Operator: Cr3000 per month. Requires at least Computer 1. Many civilian ships forgo this position and have the pilot operate the ship's sensors, but this limits their utility in the case of emergency or combat. For this reason, military vessels will have dedicated sensor operators. A ship can have a single Sensor Operator, though military craft often carry multiple Sensor Operators in the event of crew casualties.

Engineer: Cr4000 per month. Requires the Engineering skill at level 1. The Engineer maintains the ship and operates its drives. Any ship must have at least one engineer with Engineering 1 per 35 tons of drives and power plant. An engineer with Engineering 2 could run 70 tons of drives and power plants. Ships with less than 35 tons of drive and power plant do not require a full-time engineer. On larger ships, the engineer may have assistants, who are also required to have at least Engineering-1 so that there is one level of Engineering per 35 tons of drives and power plant.

Medic: Cr3000 per month. Requires at least Medicine 1. At Medicine 2 or better, the Medic is referred to as the Surgeon. The Medic provides medical care to passengers and crew and oversees any Low Berths. Any legitimate ship carrying passengers requires a Medic, and many ships without passengers employ one to care for the crew. Each level of the Medicine skill allows the Medic to care for 50 crew and passengers.

Purser: Cr3000 per month. Requires at least Administration 1. The Purser manages the ship's accounts and supplies. This position is often combined with another job, usually the Steward, with no additional pay. Smaller ships with a single Steward rarely have a Purser.

Gunner: Cr1000 per month. Requires at least Gunnery 1. Any armed ship requires one gunner per turret unless the guns are automated.

Security and Marines: Cr1000 per month. Requires at least Gun Combat 1. This position is optional. Security personnel protect a civilian starship against criminals and pirates. They are most often called upon to handle rowdy passengers. Military ships often carry marines for boarding actions and planetary operations.

Steward: Cr1000 per month. Requires at least Steward 1. Any ship carrying passengers requires one or more Stewards to care for passengers., as described later in this chapter.

FUEL

Interstellar jumps require **liquid hydrogen fuel**. Starports rated A or B sell refined fuel at Cr500 per ton. Starports rated A to D sell unrefined fuel at Cr100 per ton. Military or exploration ships usually have on-board fuel processors capable of refining fuel; small commercial ships often avoid using refineries to save cargo space and because the purification process is too long to be economically viable.

Ships may refuel, free of charge, by two other means:

A ship can **draw water** from any world with a hydrographic rating of 1 or higher. Fuel extracted in this manner is unrefined. Most worlds with starports discourage or ban this practice, but it is a common practice on the frontier. Consider water drawn in this manner Unrefined Fuel. Drawing water takes 1 hour per 20 tons of water.

A ship can **scoop** gas giants for hydrogen fuel. Fuel gathered in this manner is unrefined. This takes 1D hours per 40 tons of fuel. Note that gas giants are a favorite ambush spot for pirates. Fuel scooped from a gas giant is Unrefined.

Fuel processors, if installed, can refine unrefined fuel into refined fuel. A ton of fuel processing equipment can refine 5 tons of unrefined fuel into refined fuel per day.

LIFE SUPPORT

Each person, passenger, or crew incurs a cost of Cr2000 per month for life support, other expenses, and overhead. Luxury life support—gourmet food and other luxurious amenities—costs Cr4000 per month. Note that High Passage does not require luxury life support. However, advertising luxury life support grants an additional die to the High Passenger availability roll at Starports rated C or better on the Available Freight and Passengers Table below.

Each occupied Low passenger berth costs Cr100 per month for life support in standard travel. Low berth occupants can survive much longer – decades or more – without external life support by putting the passenger into Deep Freeze, but that imposes DM-2 penalty on their revival throws.

Purchasing life support in bulk reduces costs and allows the ship to operate for a longer time away from port. A ton of life support supplies – enough for 20 persons for a month – costs Cr34000 instead of Cr40000.

PORT FEES

A starport berth costs Cr100 for 6 days, and Cr100 per day afterwards.

MAINTENANCE

Ships require maintenance work at a starport rated C or better once per year. This costs 0.1% of the ship's listed price. If the ship skips maintenance, the Engineer must throw Engineering/INT 8+ each month to avoid a mishap; each mishap means that a single ship system has broken down: roll a single combat Hit and apply the damage to the vessel. Apply a DM-1 for every additional month of maintenance skipped. However, reroll any "Critical Hit" results rolled in this case: ships do not explode due to faulty maintenance. At least, not normally.

STARSHIP REVENUE

Ships generate revenue by carrying passengers, freight, mail, or by hiring out as charters. Speculative trade, described in the Speculative Trade chapter, can also generate revenue.

PASSAGE

There are five types of ship passage tickets. Prices are per parsec traveled. For example, traveling two parsecs by Jump-2 by Middle Passage would cost Cr16000 and take one week. The same ticket bought on a Jump-1 ship would cost the same but take 2 weeks minimum.

Roll on the **Available Freight and Passengers** table below for available passengers of each class going to a destination world. A ship may remain in port to look for more passengers; roll again every 3 days after the first six, but they must now use the new number of passengers, even if that number is lower than originally rolled. A ship can take on all or part of the available passengers. Roll an additional die for High Passengers if luxury life support is available.

High Passage is first-class travel with excellent service and cuisine, and costs Cr10000 per parsec travelled. The baggage allowance is 1000kg. The ship's Steward can serve 3 High Passengers during the journey per level of the Steward skill. Steward-0 does not count for the purposes of tending to High Passengers. For example, Steward-2 allows the steward to care for up to 6 High Passengers. It is normal for a ship to "bump" a Middle Passenger or two Steerage Passengers in favor of a late-arriving High Passenger. In this case, the bumped passenger's ticket is refunded, but there is no further compensation.

Middle Passage is standard travel with normal service and cuisine, and costs Cr8000 per parsec travelled. The baggage allowance for Middle Passage is 100kg. The Steward may serve 6 Middle Passengers for each level of Steward skill. Steward-0 does not count towards tending Middle Passengers.

Steerage Passage is economy-class travel at double occupancy (2 passengers per stateroom) with simple self-service cuisine. Steerage costs Cr3000 per parsec traveled. The baggage allowance for Steerage is 20kg. Each level of the Steward skill, including Steward-0, allows the steward to serve 12 Steerage passengers.

Low Passage means traveling in cryo-sleep. This is unsafe, but cheap, and costs Cr1000 per parsec traveled. The baggage allowance for Low Passage is 10kg. Cryo-sleep is dangerous and potentially lethal. The ship's Medic throws Medicine/EDU 6+ to safely thaw and revive Low Passengers. Apply the passenger's END DM to this throw as well. Failure means that the passenger dies from cryogenic shock. In the absence of a medic, the low berth (cryotube) itself can revive its occupant with an effective Medicine skill of 0 and an INT modifier of +0. A natural throw of 2 always kills the Low Passenger, regardless of the Medic's abilities or the passenger's END.

A ship's captain may hire an individual to fill a vacant position on a starship by arranging **Working Passage**. This provides the passenger with free travel, accommodations, and food in exchange for working at that shipboard post. The baggage allowance is 1000kg. Labor laws often require that the captain hire the passenger on as crew at full salary if they continue to work for more than three jumps.

Stowing Away on a starship is ill-advised. If caught and handed over to the authorities, the stowaway will go to prison. Some captains avoid the hassle of dealing with planetary legal bureaucracies by throwing the stowaway out of the airlock, or selling them to slavers or organ-leggers.

Note that each level of Steward skill can serve either 3 High Passengers, or 6 Middle Passengers, in addition to 12 Steerage passengers. For determining the maximum number of passengers, a steward can tend to, consider each level of Steward skill as a point: each point can be allocated to serve only one class of passengers.

For example, a steward with the Steward-1 skill can serve 3 High Passengers, or 6 Middle Passengers, or 24 Steerage Passengers (since Steward 0 can cover 12 Steerage already). That same steward could not manage 2 High Passengers and 2 Middle Passengers. A steward with Steward-2 may serve 6 High Passengers, or 12 Middle Passengers, or 36 Steerage Passengers – or 3 High Passengers, 6 Middle Passengers and 12 Steerage passengers, or 3 High Passengers, and 24 Steerage Passengers, and so on.

The above availability numbers assume a tramp freighter with no fixed schedule. The Referee should allow DMs if the ship sets up a regular schedule between worlds. Rates are in Credits per Parsec travelled.

Available Freight and Passengers Table

Destination Starport Class	Freight (tons)	High Passengers (Cr10000/psc)	Mid Passengers (Cr8000/psc)	Steerage Passengers (Cr3000/psc)	Low Passengers (Cr1000/psc)
A	3Dx10	3D	3D	4D	3Dx3
B	3Dx5	2D	3D	4D	3Dx3
C	3Dx2	1D	2D	3D	3D
D	3D	0	1D	2D	2D
E	1D	0	1D/2*	1D	1D
X	0	0	0	0	0

*Round down

Bulk Freight

A ship can transport cargo in bulk. This pays Cr1000 per ton of cargo per parsec jumped. For example, transporting a ton of cargo in a single Jump-2 pays Cr2000. Roll on the Available Freight and Passengers table above for available freight going to a given destination world. A ship may remain in port to look for more cargo; roll again every 3 days. A ship can take all or part of the available cargo. A ship may break up the available freight but must take the cargo in lots of each die rolled. For example, if the ship is at a Class B starport, there is 3Dx5 tons of freight available. The Referee rolls the 3D getting a 5, 3, and 6 – therefore, there are 3 available lots of freight 25 tons, 15 tons, and 30 tons. A ship's Purser may select any or all of the lots but must take them in the full tonnage of each lot.

Mail and Incidentals

An armed merchant can sign up to carry 5 tons of mail. This pays Cr25000 per parsec. Throw 1D-1 for the actual tonnage of mail carried (minimum 1 ton).

Other ships may carry private messages and parcels to a specific location. This pays Cr20 to Cr120 (Referee's discretion) per message or parcel. As a rule of a thumb, one ton of cargo can contain up to 50 typical parcels. Messages are electronic, and the ship's computer may hold an effectively limitless number of them.

Charters

A private individual or concern may charter a ship. Payment for in-system charters is Cr4 per ton of the ship per hour, with a 12-hour minimum. For example, a 100-ton scout would cost Cr400 per hour, minimum Cr4800. Payment for interstellar charters is in 2-week blocks at Cr3000 per ton of cargo hold plus Cr24000 per passage stateroom and Cr3000 per low berth. The owner of the vessel still pays the overhead and supplies a crew. For example, a 100-ton Scout has four staterooms (Cr96000), one low berth (Cr3000), and 14 tons of cargo (Cr56000) for a total potential of Cr155000 for a charter of two weeks. Details of possible charters are left to the Referee.

Electronics and Sensors

All starships have a computer and assorted electronics and avionics. The computer model determines two things. The first is the highest jump rating the ship can achieve, and the other is a DM to all sensor throws, as shown in the following table.

Throw Computer/INT 8+ to use a ship's sensors for scanning and detection purposes. Apply the ship's Sensor DM to this throw. The Referee may determine various further DMs to this throw based on the task at hand.

Starship sensors automatically detect any civilian ship present in the local planetary system if it is using a transponder. Smugglers, pirates, and naval vessels often turn off such transponders and thus require a Computer/INT 8+ throw to detect at long range. A stealthed ship applies DM-4 to detection throws. Detecting small craft applies a further DM-2. Detecting ships at short range is done at DM+2.

Sensors are also useful in ship combat, as described in the Space Combat chapter.

Computers and Electronics

Computer	TL	Max. Jump Rating	Sensor DM
Baseline	-	-	DM-4
Model/1	7	Jump-1	DM-2
Model/2	9	Jump-2	DM+0
Model/3	11	Jump-3	DM+1
Model/4	12	Jump-4	DM+2
Model/5	13	Jump-5	DM+2
Model/6	14	Jump-6	DM+3
Model/7	15	Jump-6	DM+4
Model/8	16	Jump-6	DM+5

